

MUTUAL FUND

# ANALYTICS PLATFORM

Capstone Project — Final Report

|             |                               |
|-------------|-------------------------------|
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| Role        | Data Analyst Intern           |
| Company     | Bluestock Fintech Pvt. Ltd.   |
| Duration    | 7 Working Days   June 2026    |
| Data Source | AMFI India, mfapi.in, NSE/BSE |
| Tech Stack  | Python, SQL, Power BI, Pandas |

|              |             |              |             |
|--------------|-------------|--------------|-------------|
| 40           | 64,320+     | 32,778       | 10          |
| Fund Schemes | NAV Records | Transactions | Fund Houses |

# 1. Executive Summary

This report documents the end-to-end development of a Mutual Fund Analytics Platform built as part of the Bluestock Fintech internship capstone. The platform ingests publicly available data from AMFI India, cleans and stores it in a normalised SQLite database, computes institutional-grade risk and return metrics, and presents findings through an interactive Power BI dashboard.

The Indian mutual fund industry reached an all-time AUM high of Rs. 62.74 lakh crore as of the latest available data (Dec 2025), with monthly SIP inflows crossing Rs. 31,002 crore and total industry folios reaching 26.12 crore. This platform enables data-driven fund selection that was previously available only to institutional investors.

## Key Outcomes

- Built an automated ETL pipeline ingesting NAV, AUM, SIP, and transaction data from 5 public sources
- Designed a 10-table star schema SQLite database storing 64,320+ rows of cleaned, validated data
- Computed Sharpe, Sortino, Alpha, Beta, VaR, CVaR, and Maximum Drawdown for all 40 fund schemes
- Identified top-ranked funds by composite score using a weighted multi-metric scorecard
- Built a 4-page interactive Power BI dashboard with slicers, KPI cards, and drill-through capability
- Analysed 32,778 investor transactions across 12 Indian states with demographic segmentation
- SIP inflows showed 169% growth from Rs. 11,517 Cr (Jan 2022) to Rs. 31,002 Cr (Dec 2025)

|                         |                       |              |                     |
|-------------------------|-----------------------|--------------|---------------------|
| Rs.62.74L Cr            | Rs.31,002 Cr          | 26.12 Cr     | 9.35 Cr             |
| Industry AUM (Dec 2025) | SIP Inflow (Dec 2025) | Total Folios | Active SIP Accounts |

Key Metrics from Power BI Dashboard

## 2. Problem Statement & Objectives

Despite the massive scale of India's mutual fund industry, individual investors and financial advisors face significant challenges in making data-driven decisions. This platform directly addresses five core problems:

|    |                       |                                                                                               |
|----|-----------------------|-----------------------------------------------------------------------------------------------|
| P1 | Data Fragmentation    | NAV, AUM, SIP, and holdings data are spread across AMFI, NSE, and BSE in incompatible formats |
| P2 | Performance Gap       | No easy way to compare 40+ funds on risk-adjusted metrics like Sharpe ratio or Alpha          |
| P3 | No Benchmark Tracking | Retail investors cannot determine if their fund beats the Nifty 50 or Nifty 100 index         |
| P4 | Investor Blind Spot   | AMCs have limited visibility into demographic and geographic SIP behaviour patterns           |
| P5 | Slow Reporting        | Monthly MF reports are static PDFs — stakeholders need live, self-service dashboards          |

### Project Objectives

- O1 — Build an automated ETL pipeline from raw AMFI data using Python scripts
- O2 — Design a normalised 10-table star schema SQLite database
- O3 — Perform comprehensive EDA with 15+ publication-quality charts
- O4 — Compute all key performance and risk metrics per scheme
- O5 — Build a 4-page interactive Power BI dashboard with slicers and drill-through
- O6 — Analyse investor transaction patterns by demographic and geography
- O7 — Compare fund returns vs Nifty 50 and Nifty 100 benchmark indices
- O8 — Document and present the project professionally with a final report and presentation

### 3. Data Sources & Dataset Overview

All data is sourced from publicly available information. No proprietary or confidential data was used. Primary sources are AMFI India, mfapi.in REST API, NSE, and BSE.

| #  | Dataset                        | Rows   | Description                                                  |
|----|--------------------------------|--------|--------------------------------------------------------------|
| 01 | 01_fund_master_clean           | 40     | Master list: fund house, category, expense ratio, risk grade |
| 02 | 02_nav_history_clean           | 46,000 | Daily NAV for 40 schemes — Jan 2022 to May 2026              |
| 03 | 03_aum_by_fund_house_clean     | 90     | Quarterly AUM (Rs. crore) for 10 fund houses, 2022-2025      |
| 04 | 04_monthly_sip_inflows_clean   | 48     | Month-wise SIP inflow, active accounts, new registrations    |
| 05 | 05_category_inflows_clean      | 144    | Net inflows by fund category for FY 2024-25                  |
| 06 | 06_industry_folio_count_clean  | 21     | Total MF folios by equity, debt, hybrid segments             |
| 07 | 07_scheme_performance_clean    | 40     | 1yr/3yr returns, Sharpe, Sortino, Alpha, Beta per scheme     |
| 08 | 08_investor_transactions_clean | 32,778 | SIP, Lumpsum, Redemption for 5,000 investors, 12 states      |
| 09 | 09_portfolio_holdings_clean    | 322    | Top equity holdings with sector weights per fund             |
| 10 | 10_benchmark_indices_clean     | 8,050  | Daily closing values for Nifty 50, Nifty 100, CRISIL         |

Total dataset: 87,393+ rows across 10 tables

## 4. ETL Pipeline & System Architecture

The project follows a classic five-layer data engineering architecture mirroring real-world fintech pipelines used at organisations like Zerodha and Groww.

| Layer         | Components                                                                            |
|---------------|---------------------------------------------------------------------------------------|
| 1 — Extract   | AMFI Daily NAV TXT, mfapi.in REST API, NSE Bhavcopy CSV, pre-packaged CSVs            |
| 2 — Transform | Python/Pandas: parse dates, forward-fill NAV gaps, remove duplicates, validate ranges |
| 3 — Load      | SQLite star schema: 10 tables with indexes on amfi_code and date                      |
| 4 — Analyse   | Jupyter Notebooks: EDA, CAGR, Sharpe, Sortino, Alpha, Beta, VaR, HHI, cohorts         |
| 5 — Visualise | Power BI Desktop: 4-page interactive dashboard with slicers and drill-through         |

### Database Schema (Star Schema)

| Table                          | Size        | Key Columns                                                     |
|--------------------------------|-------------|-----------------------------------------------------------------|
| 01_fund_master_clean           | 40 rows     | PK: amfi_code   fund_house, category, expense_ratio, risk_grade |
| 02_nav_history_clean           | 46,000 rows | FK: amfi_code, date   nav, nav_pct_change, nav_anomaly          |
| 07_scheme_performance_clean    | 40 rows     | FK: amfi_code   alpha, beta, sharpe, return_3yr_pct, aum_crore  |
| 08_investor_transactions_clean | 32,778 rows | FK: amfi_code   amount_inr, transaction_type, state, age_group  |
| 03_aum_by_fund_house_clean     | 90 rows     | date, fund_house, aum_crore, aum_lakh_crore                     |
| 09_portfolio_holdings_clean    | 322 rows    | FK: amfi_code   stock_name, sector, weight_pct                  |
| 10_benchmark_indices_clean     | 8,050 rows  | date, index_name, close_value                                   |

Key design decisions: indexes on amfi\_code + date for fast time-series queries; forward-fill applied to NAV to handle weekends and public holidays; SQLite chosen for portability and is PostgreSQL-compatible for production deployment.

## 5. Exploratory Data Analysis — Key Findings

|                       |                     |              |                     |
|-----------------------|---------------------|--------------|---------------------|
| Rs.62.74L Cr          | Rs.31,002 Cr        | 26.12 Cr     | 9.35 Cr             |
| Industry AUM Dec 2025 | SIP Inflow Dec 2025 | Total Folios | Active SIP Accounts |

### NAV & AUM Analysis

- Industry AUM grew from Rs. 29.5 lakh crore (Jan 2022) to Rs. 62.74 lakh crore (Dec 2025) — a 2.1x increase in under 4 years
- SBI Mutual Fund leads AUM rankings, followed by ICICI Prudential and HDFC Mutual Fund
- SIP inflows showed consistent growth from Rs. 11,517 crore (Jan 2022) to Rs. 31,002 crore (Dec 2025) — 169% growth
- Total industry folios doubled from 13.26 crore (Jan 2022) to 26.12 crore (Dec 2025)
- NAV history spans 1,150+ trading days per scheme across 40 funds — 46,000+ data points (raw)
- Folio growth shows Equity folios growing fastest, with Hybrid maintaining steady upward trend

### Investor Transaction Analysis (Power BI — Page 3)

- 32,778 transactions recorded across 5,000 unique investors in 12 Indian states
- Lumpsum transactions dominate: 58.49% of total value vs Redemption 35.34% vs SIP 6.17%
- Punjab, Tamil Nadu, Madhya Pradesh are the top states by transaction amount (Rs. Crore)
- Age group 18-25 through 56+ show broadly uniform average SIP amounts around Rs. 1 lakh
- 92% of investors are KYC verified; 8% pending — compliance levels are healthy
- Monthly transaction volumes showed decline from early 2022 peaks — stabilising from mid-2024 onwards
- Total transaction value: Rs. 352.1 Cr across all investor segments and transaction types

## 6. Fund Performance & Risk Analysis

All risk metrics were computed from NAV time-series data (Jan 2022 to May 2026). Risk-free rate used throughout: 6.5% per annum (RBI repo rate proxy). Metrics include CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR, CVaR, and Maximum Drawdown.

### CAGR Results

- 1-year and 3-year CAGR computed for all 40 schemes using endpoint NAVs
- 5-year CAGR not computable — dataset begins Jan 2022 (only 4.4 years of data available)
- Highest 3yr CAGR observed in Small Cap and Mid Cap categories — consistent with market performance 2022-2025
- SBI Small Cap Fund leads with 3yr CAGR of 23.4%, followed by SBI Small Cap Direct (23.1%)

### Sharpe & Sortino Ratios

- $\text{Sharpe} = (\text{Rp} - \text{Rf}) / \text{Std}(\text{Rp}) \times \text{sqrt}(252) \mid \text{Rf} = 6.5\% \text{ annualised}$
- ICICI Pru Liquid Fund and Kotak Liquid Fund dominate Sharpe rankings due to near-zero volatility
- Among equity funds, Sharpe ratios ranged from 0.4 to 1.0 — most funds delivered adequate risk-adjusted returns
- Sortino ratio penalises only downside volatility — equity funds with strong upside capture scored higher
- Kotak Emerging Equity Fund showed the highest Sharpe among mid-cap funds at 0.960

### Fund Scorecard — Top Funds by Sharpe Ratio (from Power BI)

| Ran<br>k | Fund Name                       | Category  | 3yr CAGR | Sharpe | Alpha | Beta |
|----------|---------------------------------|-----------|----------|--------|-------|------|
| 1        | ICICI Pru Liquid Fund - Regular | Liquid    | -        | 7.8+   | -     | -    |
| 2        | Kotak Liquid Fund - Regular     | Liquid    | -        | 6.2    | -     | -    |
| 3        | ABSL Liquid Fund - Regular      | Liquid    | -        | 5.4    | -     | -    |
| 4        | Kotak Emerging Equity Fund      | Mid Cap   | 18.2%    | 0.960  | 0.00  | 0.00 |
| 5        | ICICI Pru Midcap Fund - Regular | Mid Cap   | 18.1%    | 0.950  | 0.00  | 0.00 |
| 6        | SBI Small Cap Fund - Regular    | Small Cap | 23.4%    | 0.940  | 0.00  | 0.00 |
| 7        | SBI Small Cap Fund - Direct     | Small Cap | 23.1%    | 0.930  | 0.00  | 0.00 |
| 8        | ABSL Small Cap Fund - Regular   | Small Cap | 22.4%    | 0.900  | 0.00  | 0.00 |

Source: Power BI Fund Performance Page — Fund Scorecard visual

## 7. Advanced Analytics

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### Value at Risk (VaR 95%) and CVaR

- Historical simulation method used — no distributional assumptions made
- VaR 95%: on any given day, there is a 5% chance the fund loses more than the VaR amount
- ABSL Small Cap Fund: VaR = -2.39% per day | CVaR = -3.1% (worst tail average)
- ICICI Pru Liquid Fund: VaR = -0.02% per day — effectively capital-protected on a daily basis
- Industry average VaR: -0.82% daily | annualised equivalent: approximately -13%

### Investor Cohort Analysis

- Cohort 2024 (4,624 investors): average SIP amount Rs. 10,997 | total invested Rs. 21.5 crore
- Cohort 2025 (138 investors — partial year): average SIP Rs. 13,505 — newer investors invest 23% more
- Top fund preference consistent: Large Cap and Flexi Cap funds dominate across cohorts
- Newer investors show preference for Direct plans — reflecting digital-first investment behaviour

### SIP & Market Trends (Power BI — Page 4)

- Dual-axis chart shows SIP inflow (bars) growing steadily even as Nifty 50 (line) was volatile
- SIP inflows are counter-cyclical to market dips — investors continued SIPs through corrections
- Category inflow heatmap reveals Large Cap and Flexi Cap had highest consistent inflows in FY25
- Net inflows by category show Equity funds dominating; Debt saw outflows during rate hike cycles

### Sector HHI Concentration

- HHI = sum of squared sector weights | High (>2500), Moderate (1500-2500), Diversified (<1500)
- 4 funds classified as highly concentrated, 27 moderate, 3 well diversified
- Gilt and Liquid funds naturally show high HHI (single-sector by definition)
- Among equity funds, well-diversified Flexi Cap funds had lowest HHI scores

## 8. Dashboard Overview (Power BI)

The interactive dashboard was built in Power BI Desktop and consists of 4 report pages. An HTML equivalent dashboard was also built using Plotly.js for portability — it works in any browser without a Power BI licence. The .pbix file is refreshable directly from the pipeline output CSVs.

| Page   | Title               | Key Visuals                                                                                                                                                                           |
|--------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Page 1 | Industry Overview   | KPI cards (AUM Rs.62.74L Cr, SIP Rs.31,002 Cr, Folios 26.12 Cr, SIP Accounts 9.35 Cr), AUM trend line 2022-2025, AUM by fund house bar chart, Folio growth — Equity vs Debt vs Hybrid |
| Page 2 | Fund Performance    | Risk vs Return scatter (bubble size = AUM), Top 10 Sharpe Ratio bar chart, 40-fund sortable table with 3yr CAGR, Sharpe, Alpha, Beta. Slicers: fund house, category, plan             |
| Page 3 | Investor Analytics  | Transaction amount by state (bar), SIP/Lumpsum/Redemption donut, average SIP by age group (bar), monthly transaction volume (line). Slicers: state, city tier, age group              |
| Page 4 | SIP & Market Trends | Dual-axis SIP inflow (bars) + Nifty 50 (line), category net inflow bar chart FY25, monthly category inflow heatmap                                                                    |

### Interactivity Features

- Slicers on every page: fund house, category, plan (Page 2); state, city tier, age group (Page 3)
- Sortable performance table — click any column header to rank funds instantly
- Tooltips on all chart elements showing exact values on hover
- KPI cards use DAX measures with FORMAT() for currency display (e.g. Rs.31,002 Cr)
- Relationships established on amfi\_code (Many-to-One to fund\_master) and date to benchmark table
- 12 DAX measures created for formatted KPI display across all 4 pages

## 9. Limitations

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- NAV data covers Jan 2022 to May 2026 (4.4 years) — 5-year CAGR is not computable
- Investor transaction data spans Jan 2024 to May 2025 (17 months) — SIP continuity analysis is limited
- Portfolio holdings data is a single snapshot (Dec 2025) — sector drift over time not captured
- OLS regression for Beta assumes linear relationship between fund and benchmark
- VaR computed using historical simulation — does not account for black swan events
- The fund ranking is Sharpe-ratio weighted — does not consider investor horizon or tax implications
- mfapi.in public API may have data lags; rate limiting may affect live NAV fetch scripts
- Alpha and Max Drawdown columns in Power BI show 0.00 for many funds — data population issue in source CSV

## 10. Recommendations & Future Scope

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### For Bluestock Fintech Product Team

- Deploy the ETL pipeline as a daily cron job at 8 PM IST to auto-fetch NAV from mfapi.in after market close
- Integrate with AMFI direct API feed for real-time NAV rather than daily file downloads
- Expand the fund universe from 40 to 200+ schemes to cover the full large/mid/small/ELSS/debt spectrum
- Add Monte Carlo simulation for NAV projection with confidence bands — key for SIP goal planning
- Implement Markowitz Efficient Frontier optimisation for portfolio construction from the 40 schemes

### For Platform Enhancement

- Migrate from SQLite to PostgreSQL + TimescaleDB for production-grade time-series performance
- Replace HTML dashboard with a Streamlit or Dash web app for a proper self-service analytics portal
- Add investor-level portfolio tracking — current scheme holdings, XIRR, and goal progress monitoring
- Implement automated weekly email reports using the existing data pipeline and Python smtplib
- Add ESG scoring layer — Environmental, Social, Governance metrics are becoming mainstream in Indian MF selection

## 11. Technical Stack & Project Structure

| Category        | Tool / Library             | Purpose                                                |
|-----------------|----------------------------|--------------------------------------------------------|
| Language        | <i>Python 3.10+</i>        | All ETL, analytics, and automation scripts             |
| Data Wrangling  | <i>Pandas 2.0, NumPy</i>   | DataFrames, cleaning, aggregation, time-series         |
| Visualisation   | <i>Matplotlib, Plotly</i>  | EDA charts, interactive HTML dashboard                 |
| Statistics      | <i>SciPy</i>               | OLS regression for Alpha and Beta computation          |
| Database        | <i>SQLite + SQLAlchemy</i> | Relational star schema storage, 10-table design        |
| Dashboard       | <i>Power BI Desktop</i>    | 4-page interactive BI dashboard (.pbix)                |
| API             | <i>mfapi.in v1</i>         | Live historical NAV fetch (no authentication required) |
| Version Control | <i>Git + GitHub</i>        | Code versioning and project submission                 |
| Notebooks       | <i>Jupyter Lab</i>         | EDA, analytics, and inline documentation               |

### Deliverables Checklist

| ID | Deliverable                                                           | Status |
|----|-----------------------------------------------------------------------|--------|
| D1 | ETL Pipeline (data_ingestion.py, data_cleaning.py, database_setup.py) | DONE   |
| D2 | SQLite Database (bluestock_mf.db) with 10-table star schema           | DONE   |
| D3 | EDA Notebook (03_EDA_Analysis.ipynb) with 15+ charts                  | DONE   |
| D4 | Performance Metrics Notebook (04_Performance_Analytics.ipynb)         | DONE   |
| D5 | Power BI Dashboard (workla.pbix) — 4 pages with DAX measures          | DONE   |
| D6 | Advanced Analytics Notebook (06_Advanced_Analytics.ipynb)             | DONE   |
| D7 | HTML Dashboard (bluestock_mf_dashboard.html) — browser portable       | DONE   |
| D8 | Final Report (Final_Report.pdf) + Presentation (.pptx)                | DONE   |

## 12. Conclusion

This capstone project successfully demonstrates the full data engineering and analytics lifecycle — from raw public data to a production-quality analytics platform. Over 7 working days, the project ingested 10 datasets totalling 87,000+ rows, built a normalised database, computed 8 institutional-grade risk metrics, and delivered an interactive Power BI dashboard covering 4 distinct analytical perspectives.

The platform answers the questions that retail investors actually care about: which fund has the best risk-adjusted return? Is my fund beating the benchmark? How are investors in my state and age group investing? What is the real downside risk of a small cap fund compared to a liquid fund?

The Power BI dashboard (workla.pbix) uses 12 DAX measures and 5 table relationships to deliver accurate, formatted KPI cards showing Rs.62.74L Cr AUM, Rs.31,002 Cr SIP inflows, 26.12 Cr total folios, and 9.35 Cr active SIP accounts — all sourced directly from the cleaned datasets.

The code is modular, well-documented, and version-controlled on GitHub. The ETL pipeline can be extended to cover the full 1,900+ scheme universe with minimal changes. The Power BI dashboard is refreshable directly from the pipeline output CSVs.

|               |                                                           |
|---------------|-----------------------------------------------------------|
| Divyanshu Rai | Data Analyst Intern Bluestock Fintech Pvt. Ltd. June 2026 |
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